Lesson 7

Joins

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Joining Related Tables

- With normalized databases, information required for a single result set may be located in more than one table

- When defining a JOIN, you must:
  - Define the tables to be joined
  - Define the join type
  - Define a join condition, which is made up of the columns on which the tables are joined and the logical operator

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INNER JOIN

- INNER JOINs allow you to match related records from different tables
- Return only the rows from tables that match the join condition
- INNER keyword is optional

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Student

	student_id	first_name	last_name	teacher
1	1	s1	ss1	1
2	2	s2	ss2	2
3	3	s3	ss3	2
4	4	s4	ss4	1
5	5	s5	ss5	NULL
6	6	s6	ss6	2
7	7	s7	ss7	2
8	8	d8	ss8	1

Teacher

	teacher_id	first_name	last_name
1	1	t1	tt1
2	2	t2	tt2
3	3	t3	tt3

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To generate a class list showing each student's name and their teacher

```
SELECT
      student.first_name,
      student.last_name,
      teacher.first_name,
      teacher.last_name
FROM
      student
INNER JOIN
      teacher
ON
      student.teacher = teacher.teacher_id
```

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	first_name	last_name	first_name	last_name
1	s1	ss1	t1	tt1
2	s2	ss2	t2	tt2
3	s3	ss3	t2	tt2
4	s4	ss4	t1	tt1
5	s6	ss6	t2	tt2
6	s7	ss7	t2	tt2
7	d8	ss8	t1	tt1

Student 5 and Teacher 3 are missing!!!

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Same example using alias

```
SELECT
  s.first_name      AS 'Student FN',
  s.last_name AS 'Student LN',
  t.first_name      AS 'Teacher FN',
  t.last_name AS 'Teacher LN'
FROM
  student AS s
INNER JOIN
  teacher AS t
ON
  s.teacher = t.teacher_id
```

	Student FN	Student LN	Teacher FN	Teacher LN
1	s1	ss1	t1	tt1
2	s2	ss2	t2	tt2
3	s3	ss3	t2	tt2
4	s4	ss4	t1	tt1
5	s6	ss6	t2	tt2
6	s7	ss7	t2	tt2
7	d8	ss8	t1	tt1

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LEFT and RIGHT OUTER JOIN

- OUTER JOIN also include records from one or both tables that do not have corresponding record(s) in the other table
- Returns all rows from one table (Left or Right) and only information from rows that are in common from other table
- OUTER keyword is optional

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```sql
SELECT
  s.first_name AS 'Student FN',
  s.last_name  AS 'Student LN',
  t.first_name AS 'Teacher FN',
  t.last_name  AS 'Teacher LN'
FROM
  student AS s
LEFT OUTER JOIN
  teacher AS t
ON
  s.teacher = t.teacher_id
```

	Student FN	Student LN	Teacher FN	Teacher LN
1	s1	ss1	t1	tt1
2	s2	ss2	t2	tt2
3	s3	ss3	t2	tt2
4	s4	ss4	t1	tt1
5	s5	ss5	NULL	NULL
6	s6	ss6	t2	tt2
7	s7	ss7	t2	tt2
8	d8	ss8	t1	tt1

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Right outer join

```
SELECT
      s.first_name AS 'Student FN',
      s.last_name  AS 'Student LN',
      t.first_name AS 'Teacher FN',
      t.last_name  AS 'Teacher LN'
FROM
      teacher AS t
Right OUTER JOIN
      student AS s
ON
      s.teacher = t.teacher_id
```

Tables swapped

This example and previous one are the same

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```sql
SELECT
        s.first_name AS 'Student FN',
        s.last_name AS 'Student LN',
        t.first_name AS 'Teacher FN',
        t.last_name AS 'Teacher LN'
FROM
        student AS s
RIGHT OUTER JOIN
        teacher AS t
ON
        s.teacher = t.teacher_id
```

	Student FN	Student LN	Teacher FN	Teacher LN
1	s1	ss1	t1	tt1
2	s4	ss4	t1	tt1
3	d8	ss8	t1	tt1
4	s2	ss2	t2	tt2
5	s3	ss3	t2	tt2
6	s6	ss6	t2	tt2
7	s7	ss7	t2	tt2
8	NULL	NULL	t3	tt3

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FULL OUTER JOIN

- The FULL OUTER JOIN is a combination of the LEFT JOIN and the RIGHT JOIN
- A FULL OUTER JOIN displays every row from every table in the JOIN clause
- This can be helpful in finding unmatched rows when relational integrity is not being enforced on the tables.

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```sql
SELECT
  s.first_name    AS 'Student FN',
  s.last_name     AS 'Student LN',
  t.first_name    AS 'Teacher FN',
  t.last_name     AS 'Teacher LN'
FROM
  student AS s
FULL OUTER JOIN
  teacher AS t
ON
  s.teacher = t.teacher_id
```

	Student FN	Student LN	Teacher FN	Teacher LN
1	s1	ss1	t1	tt1
2	s2	ss2	t2	tt2
3	s3	ss3	t2	tt2
4	s4	ss4	t1	tt1
5	s5	ss5	NULL	NULL
6	s6	ss6	t2	tt2
7	s7	ss7	t2	tt2
8	d8	ss8	t1	tt1
9	NULL	NULL	t3	tt3

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Cross JOIN

- Cross joins provide a Cartesian product of the two tables. Every row from the first table is joined with every row from the second table

- This JOIN does not have a ON clause

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```
SELECT
  s.first_name,
  s.last_name,
  t.first_name,
  t.last_name
FROM
  student AS s
CROSS JOIN
  teacher AS t
```

	Student FN	Student LN	Teacher FN	Teacher LN
1	s1	ss1	t1	tt1
2	s2	ss2	t1	tt1
3	s3	ss3	t1	tt1
4	s4	ss4	t1	tt1
5	s5	ss5	t1	tt1
6	s6	ss6	t1	tt1
7	s7	ss7	t1	tt1
8	d8	ss8	t1	tt1
9	s1	ss1	t2	tt2
10	s2	ss2	t2	tt2
11	s3	ss3	t2	tt2
12	s4	ss4	t2	tt2
13	s5	ss5	t2	tt2
14	s6	ss6	t2	tt2
15	s7	ss7	t2	tt2
16	d8	ss8	t2	tt2
17	s1	ss1	t3	tt3
18	s2	ss2	t3	tt3
19	s3	ss3	t3	tt3
20	s4	ss4	t3	tt3
21	s5	ss5	t3	tt3
22	s6	ss6	t3	tt3
23	s7	ss7	t3	tt3
24	d8	ss8	t3	tt3

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- To use the Nums table in TSQL2012 DB to generate a result with a row for each weekday (1 through 7) and shift number (1 through 3), assuming there are three shifts a day.

```
SELECT
      D.n AS theday, S.n AS shiftno
FROM
      dbo.Nums AS D
CROSS JOIN
      dbo.Nums AS S
WHERE
      D.n <= 7 AND S.n <= 3
ORDER BY
      theday, shiftno;
```

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Joining a table with itself

- This situation is known as a SELF-JOIN, and it occurs when you want to compare records in a table to each other

- Typical example of a self-join is a “flattened” employees table that contains information on an organizational hierarchy

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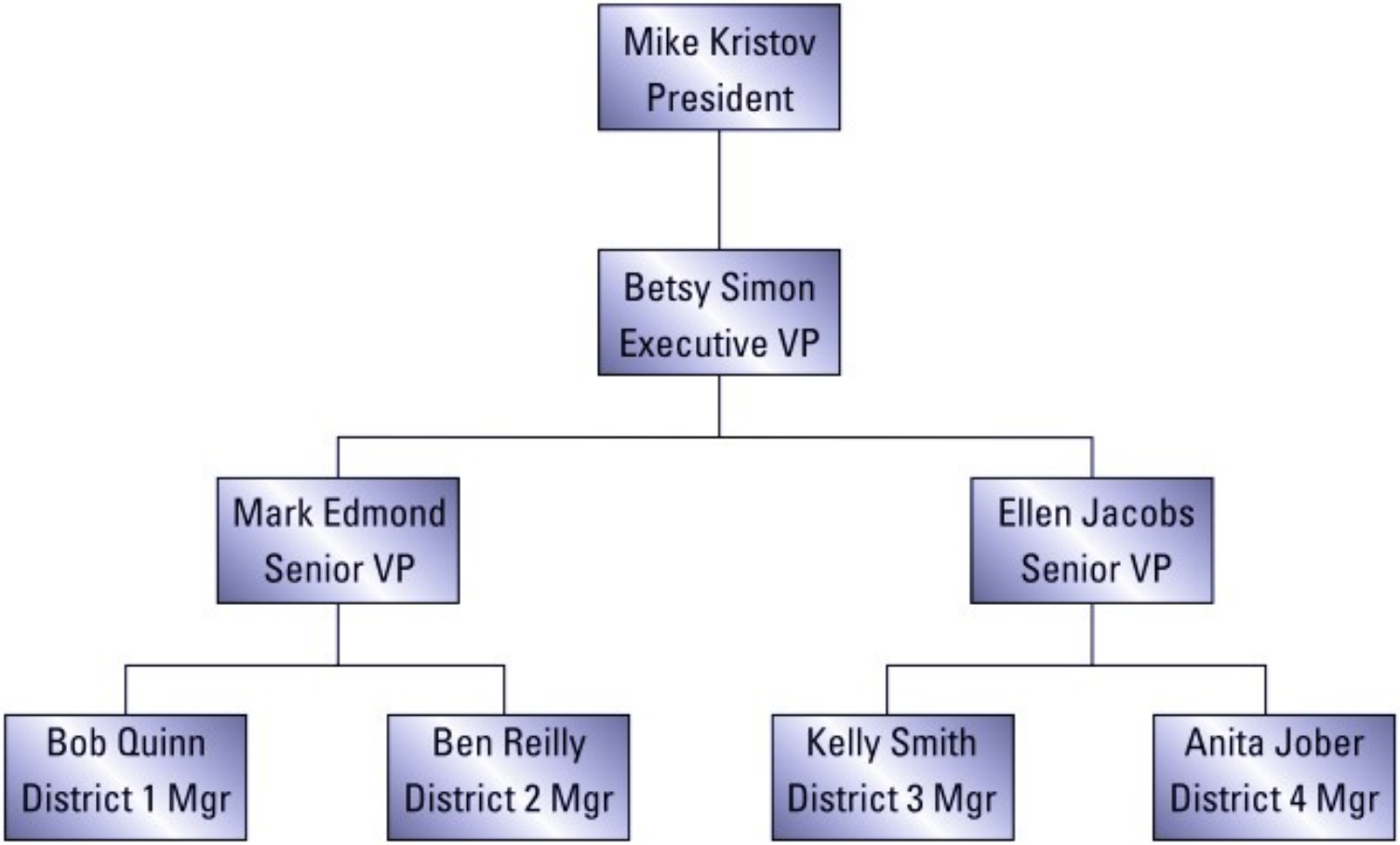
Employees Table

employee_id	*first_name*	*last_name*	*title*	*manager_id*
1	Mike	Kristov	President	NULL
2	Betsy	Simon	Executive Vice President	1
3	Mark	Edmond	Senior Vice President	2
4	Ellen	Jacobs	Senior Vice President	2
5	Bob	Quinn	District 1 Manager	3
6	Ben	Reilly	District 2 Manager	3
7	Kelly	Smith	District 3 Manager	4
8	Anita	Jober	District 4 Manager	4

Internal reference

Representation of previous table



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```sql
SELECT
  e.first_name    AS 'Employee FN',
  e.last_name     AS 'Employee LN',
  e.title    AS 'Employee Title',
  m.first_name    AS 'Manager FN',
  m.last_name     AS 'Manager LN'
FROM
  employees AS e
LEFT OUTER JOIN
  employees AS m
ON
  e.manager_id = m.employee_id
```

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```
Employee FN Employee LN Employee Title           Manager FN Manager LN
----------- ----------- ------------------------- ---------- ----------
Mike        Kristov     President                 NULL       NULL
Betsy       Simon       Executive Vice President  Mike       Kristov
Mark        Edmond      Senior Vice President     Betsy      Simon
Ellen       Jacobs      Senior Vice President     Betsy      Simon
Bob         Quinn       District 1 Manager        Mark       Edmond
Ben         Rilley      District 2 Manager        Mark       Edmond
Kelly       Smith       District 3 Manager        Ellen      Jacobs
Anita       Jober       District 4 Manager        Ellen      Jacobs

(8 row(s) affected)
```

If we used an INNER JOIN instead of a LEFT OUTER JOIN, my employee list would have omitted the company's president, who
has no manager

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Combining Datasets

- The UNION, EXCEPT and INTERSECT operators can be specified between two or more queries to provide a single result set

- EXCEPT and INTERSECT operators compare the result sets of two queries to determine what subset of rows should be included in the final result set

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UNION Operator

- The UNION operator allows you to combine the result sets created by multiple SELECT statements into a single result set
- The ALL keyword specifies that all rows, including duplicate rows, should be returned
- By default, duplicate rows are **not** returned

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- When specifying the UNION operator, both queries must return the same number of columns
- Each data type in the corresponding columns must be

```
SELECT
CONVERT (Char(8), Employees.EmployeeID)
FROM Employees
UNION
SELECT Customers.CustomerID FROM
Customers
```

String

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- The column titles are taken from the first query listed
- ORDER BY clauses should be defined at the end of the last SELECT statement
- Each SELECT statement should include its own WHERE clause

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```sql
use Northwind

SELECT
      FirstName+' '+LastName  AS 'Contact Name',
      [Northwind Traders]     AS 'Company',
      Employees.HomePhone     AS 'Phone'
FROM
      Employees
UNION
SELECT
      ContactName,
      CompanyName,
      Phone
FROM
      Customers
ORDER BY
      [Contact Name];
```

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EXCEPT and INTERSECT

- The EXCEPT operator returns all rows that exist in the table to the left of the operator that do not have matching rows in the table to the right

```
SELECT EmployeeKey
FROM DimEmployee
EXCEPT
SELECT EmployeeKey
FROM FactResellerSales
```

Employees who have never placed a reseller sales order

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- The same number of columns with compatible data types must be defined in both queries

- You may need to join the tables to achieve this

```
USE JOB_PORTAL_DB;
GO

SELECT SL.Login, SL.Full_Name
FROM [dbo].[Security_Logins] AS SL
EXCEPT
SELECT SL.Login, SL.Full_Name
FROM [dbo].[Security_Logins] AS SL
 INNER JOIN
 [dbo].[Security_Logins_Log] AS SLL
 ON SLL.Login = SL.Id;
```

Find users who have never logon to the system

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- INTERSECT returns matching rows from both queries only.

Find users who applied for a job in year of 2017

```
USE JOB_PORTAL_DB;
GO

SELECT SL.Login, SL.Full_Name
FROM [dbo].[Security_Logins] AS SL
INTERSECT
 SELECT SL.Login, SL.Full_Name
 FROM [dbo].[Security_Logins] AS SL
 INNER JOIN [dbo].[Applicant_Profiles] AS AP
 ON AP.Login = SL.Id
 INNER JOIN [dbo].[Applicant_Job_Applications] AS AJA
 ON AJA.Applicant = AP.Id
 WHERE AJA.Application_Date
 BETWEEN '2017-01-01' AND '2017-12-31'
```

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